

MMCIF (MMC/SD Interface) Register Manual

Ver 0.1

DM270

Digital Still Camera

Texas Instruments Japan LTD.

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MMC/SD INTERFACE

TMS320DM270 は、MMC/SD カード・インターフェースを内蔵しています。このモジュールの動作クロックは ARM クロックを分周したものを基準に動作します。クロックの詳細な設定はクロック・コントローラを参照してください。

MMC/SD Card Interface 特徴

- Support MMC and SD cards by conforming to both MMC and SPI protocols
- One or four-bit width data communication
- Software oriented implementation for future extensions
- MMC peripheral runs at a divided rate than the CPU clock.
- Supports SDIO card

MMCCTL (0003:0480) Offset : 0x0

Bit Name	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	RSV	RSV	RSV	RSV	RSV	PERMDX	PERMDR	RSV	DATEG1	DATEG0	SPIEN	CSEN	SPICRC	WIDTH	CMDRST	DATRST
R/W	-	-	-	-	-	R/W	R/W	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	-	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-11	RSV	Reserved Bits. Do not use.
10	PERMDX	Endian select enable when writing 0: Little Endian is selected. 1: Big Endian is selected.
9	PERMDR	Endian select Enable when reading 0: Little Endian is selected. 1: Big Endian is selected.
8	RSV	Reserved Bits. Do not use.
7-6	DATEG1-DATEG0	DAT3 Edge Detection Select 00: DAT3 edge detection is disabled. 01: DAT3 rising edge detection is enabled. 10: DAT3 falling edge detection is enabled. 11: DAT3 both edge detection is enabled.
5	SPIEN	SPI Mode Enable 0: MMC mode is selected. 1: SPI mode is selected.
4	CSEN	Card Select Enable when writing (in only SPI mode) 0: CS(DAT3) output is H (inactive). 1: CS(DAT3) output is L (active).
3	SPICRC	CRC Checking (in only SPI mode) 0: CRC checking is disabled. 1: CRC checking is enabled.
2	WIDTH	Data Bus Width (in only MMC mode) 0: Data bus has 1-bit, (only DAT0 is used). 1: Data Bus has 4-bit, (all DAT0-3 are used).
1	CMDRST	CMD Logic Reset 0: The CMD line portion is enabled. 1: The CMD line portion is disabled and in reset state.
0	DATRST	DAT Logic Reset 0: The DAT line portion is enabled. 1: The DAT line portion is disabled and in reset state.

MMCCLK (0003:0482) Offset : 0x1

Bit Name	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	RSV	RSV	RSV	RSV	RSV	RSV	RSV	CLKEN	CLKRT7	CLKRT6	CLKRT5	CLKRT4	CLKRT3	CLKRT2	CLKRT1	CLKRT0
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	1	1	1	1	1	1	1	1

Bit	Register Name	Description
15-9	RSV	Reserved Bits. Do not use.
8	CLKEN	CLK Enable

		0: CLK pin is disabled and fixed L. 1: CLK pin is enabled.
7-0	CLKRT7-CLKRT0	Clock Pin Rate This indicates divider rate of the function clock MMC clock = function clock /2*(ClockRate + 1)

MMCST0 (0003:0484) Offset : 0x2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	DATED	DRRDY	DXRDY	SPIERR	CRCRS	CRCRD	CRCWR	TOUTRS	TOUTRD	RSPDNE	BSYDNE	DATDNE
R/W	-	-	-	-	R	R	R	R	R	R	R	R	R	R	R	R
Default	-	-	-	-	0	0	1	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-12	RSV	Reserved Bits. Do not use.
11	DATED	DAT3 Edge Detect, bit is cleared when read by CPU 0: No DAT3 edge is detected. 1: DAT3 edge has been detected.
10	DRRDY	Data Receive Ready 0: MMCDRR is not ready. 1: MMCDRR is ready to be reading a newdata by CPU or DMA. At DAT logic reset(DATRST=1), sending command with Data Receive/Transmit Clear(DCLR=1) or reading MMCDRR, the DRRDY bit is clear to 0.
9	DXRDY	Data Transmit Ready 0: MMCDXR is not ready. 1: MMCDXR is ready to be writing a new data by CPU or DMA. At DAT logic reset(DATRST=1), sending command with Data Receive/Transmit Clear(DCLR=1) or writing MMCDXR, the DXRDY bit is set to 1.
8	SPIERR	Data Error (in SPI mode) 0: No data error is received. 1: Data error token has been received.
7	CRCRS	CRC error (response) 0: No CRC error is detected. 1: CRC error has been detected.
6	CRCRD	CRC error (read data) 0: No CRC error is detected. 1: CRC error has been detected.
5	CRCWR	CRC error (write data) 0: No CRC error is detected. 1: CRC error has been detected.
4	TOUTRS	Time-Out (response) 0: No time-out occurs. 1: Timed-out has occurred during waiting response.
3	TOUTRD	Time-Out (read data) 0: No time-out occurs. 1: Timed-out has occurred during waiting data.
2	RSPDNE	Command / Response Done 0: No Receiving response done. 1: Response successfully has received or command has sent without response.
1	BSYDNE	Busy Done 0: No busy releasing done. 1: Released from busy state or expected busy not detected.
0	DATDNE	Data Done 0: No data transfer done. 1: Data transfer of specified length successfully done.

Note1: The transition from 0 to 1 of each bit in MMCST0 may cause interrupt signal to the ARM. It depends on MMCIM setting, (see also MMCIM). Each status bit is cleared when reading by the ARM except DRRDY and DXRDY.

Note2: Because of independence of the command portion and the data portion, any command such as CMD0

(GO_IDLE_STATE) or CMD12 (STOP_TRANSMISSION) can be sent to the card, even if during block transfer. In this situation, the data portion will detect and wait releasing busy state only when the command sent was R1b (to be specifically, command with BSYEXP bit), otherwise it will keep on transferring data.

MMCST1 (0003:0486) Offset : 0x3

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	DAT3ST	DRFUL	DXEMP	CLKSTP	BUSY
R/W	-	-	-	-	-	-	-	-	-	-	-	R	R	R	R	R
Default	-	-	-	-	-	-	-	-	-	-	-	0	0	0	1	0

Bit	Register Name	Description
15-5	RSV	Reserved Bits. Do not use.
4	DAT3ST	DAT3 Status 0: The status of DAT3 is L level. 1: The status of DAT3 is H level.
3	DRFUL	DRRDY Data receive full 0: MMCDRR(data receive register) is not full. 1: MMCDRR(data receive register) is full.
2	DXEMP	DXRDY Data transmit empty 0: MMCDXR(data transmit register) is not empty. 1: MMCDXR(data transmit register) is empty.
1	CLKSTP	Clock Stop Status 0:CLK is active. 1:CLK is held L because of no sending data during CLKEN is 0, receive shift register is full, transmit shift register is empty.
0	BUSY	Busy 0: No busy signal is detected 1: Busy signal is detected after response with busy or during block write.

MMCIM (0003:0488) Offset : 0x4

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	EDATED	ERDFUL	EWREMP	ESPIERR	ECRCRS	ECRCRD	ECRCWR	ETOUTRS	ETOUTRD	ECMDDNE	EBSYDNE	EDATDNE
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-12	RSV	Reserved Bits. Do not use.
11	EDATED	DATED permission bit. 0: Prohibited 1: Permitted
10	ERDFUL	DRRDY permission bit. 0: Prohibited 1: Permitted
9	EWREMP	DXRDY permission bit. 0: Prohibited 1: Permitted
8	ESPIERR	SPIERR permission bit. 0: Prohibited 1: Permitted
7	ECRCRS	CRCRS permission bit. 0: Prohibited 1: Permitted
6	ECRCRD	CRCRD permission bit. 0: Prohibited 1: Permitted
5	ECRCWR	CRCWR permission bit. 0: Prohibited 1: Permitted
4	ETOUTRS	TOUTRS permission bit.

		0: Prohibited 1: Permitted
3	ETOUTRD	TOUTRD permission bit. 0: Prohibited 1: Permitted
2	ERSPDNE	RSPDNE permission bit. 0: Prohibited 1: Permitted
1	EBSYDNE	BSYDNE permission bit. 0: Prohibited 1: Permitted
0	EDATDNE	DATDNE permission bit. 0: Prohibited 1: Permitted

MMCTOR (0003:048A) Offset : 0x5

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	TOR7	TOR6	TOR5	TOR4	TOR3	TOR2	TOR1	TOR0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-8	RSV	Reserved Bits. Do not use.
7-0	TOR7-TOR0	Time-Out Count For Response (in MMC mode) 0000 0000: no time-out 0000 0001: 1 CLK clock 0000 0010: 2 CLK clocks 1111 1111: 255 CLK clocks Time-Out Count For Response (in SPI mode) 0000 0000: no time-out 0000 0001: 1 x 8 CLK clocks 0000 0010: 2 x 8 CLK clocks 1111 1111: 255 x 8 CLK clocks

MMCTOD (0003:048C) Offset : 0x6

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	TOD15	TOD14	TOD13	TOD12	TOD11	TOD10	TOD09	TOD08	TOD07	TOD06	TOD05	TOD04	TOD03	TOD02	TOD01	TOD00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	TOD15- TOD00	Time-Out Count For Data Read (in MMC mode) 0x0000: no time-out 0x0001: 1 CLK clock 0x0002: 2 CLK clocks 0xffff: 65535 CLK clocks Time-Out Count For Data Read (in SPI mode) 0x0000: no time-out 0x0001: 1 x 8 CLK clocks 0x0002: 2 x 8 CLK clocks 0xffff: 65535 x 8 CLK clocks

MMCBLEN (0003:048E) Offset : 0x7

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	BLEN11	BLEN10	BLEN9	BLEN8	BLEN7	BLEN6	BLEN5	BLEN4	BLEN3	BLEN2	BLEN1	BLEN0

R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	1	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-12	RSV	Reserved Bits. Do not use.
11-0	BLLEN11-BLEN0	Block Length Specifies the byte count of a data block. The value 0x000 is prohibited.

Note: MMCBLLEN specifies the data block length. This value must be same as CSD register settings in the card, to be precise, READ_BL_LEN when block read, or WRITE_BL_LEN when block write.
When DMAM = '1', MMCBLLEN must take multiples of 4.

MMCNBLK (0003:0490) Offset : 0x8

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	NBLK15	NBLK14	NBLK13	NBLK12	NBLK11	NBLK10	NBLK9	NBLK8	NBLK7	NBLK6	NBLK5	NBLK4	NBLK3	NBLK2	NBLK1	NBLK0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	NBLK15-NBLK0	Number of Blocks. Specifies the total number of blocks to be transferred. The value 0x0000 means infinite blocks.

MMCNBLC (0003:0492) Offset : 0x9

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	NBLC15	NBLC14	NBLC13	NBLC12	NBLC11	NBLC10	NBLC9	NBLC8	NBLC7	NBLC6	NBLC5	NBLC4	NBLC3	NBLC2	NBLC1	NBLC0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Register Name	Description
15-0	NBLC15-NBLC0	A counter for counting number of blocks to be transferred. Writing to this register is prohibited by User. (Writing function may use for counter test.)

MMCDRR (0003:0494) Offset : 0xA

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DRR15	DRR14	DRR13	DRR12	DRR11	DRR10	DRR9	DRR8	DRR7	DRR6	DRR5	DRR4	DRR3	DRR2	DRR1	DRR0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	DRR15-DRR0	This register is used for reading of receive data from card when block read. If an odd byte number of a data block is set up by MMCBLLEN, last byte is received to DRR[7:0].

MMCDXR (0003:0496) Offset : 0xB

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DXR15	DXR14	DXR13	DXR12	DXR11	DXR10	DXR9	DXR8	DXR7	DXR6	DXR5	DXR4	DXR3	DXR2	DXR1	DXR0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	DXR15- DXR0	This register is used for writing of transmit data to card when block write. If an odd byte number of a data block is set up by MMCBLLEN, last byte is transmitted from DXR[15:8].

MMCCMD (0003:0498) Offset : 0xC

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DCLR	INITCK	WDATX	STRMTP	DTRW	RSPFMT1	REFPMT0	BSYLEN	PPLEN	RSV	CMD5	CMD4	CMD3	CMD2	CMD1	CMD0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	-	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	-	0	0	0	0	0	0

Bit	Register Name	Description
15	DCLR	Data Receive/Transmit Clear 0: No clear 1: Clear

14	INITCK	Initialize Clock Before sending the command (specified in CMD), 80 cycles of dummy clock will be inserted. This is required for resetting a card after power-on. 0: No initialization clock 1: Initialization clock insertion
13	WDATX	With Data 0: No data transfer 1: With data transfer
12	STRMTP	Stream Transfer 0: Block data transfer 1: Stream data transfer
11	DTRW	Write 0: Data read 1: Data write
10-9	RSPFMT1- RSPFMT0	Response Format 00: No response 01: R1/R4/R5/R6 (48-bit length) 10: R2 (136-bit length) 11: R3 (48-bit length, without CRC)
8	BSYEXP	Busy Expected 0: No busy signal expected. 1: Busy signal expected or in case of transfer abort.
7	PPLEN	Push Pull Enable 0: Push Pull Driver of CMD line is disable. 1: Push Pull Driver of CMD line is enable.
6	RSV	Reserved Bits. Do not use.
5-0	CMD5-CMD0	Command Index The command to be sent to the CMD line.

Note: The command to be sent to the CMD line are stored in MMCCMD. To write to MMCCMD triggers to start sending a command. Thus, the argument registers (MMCARGx) have to be written properly before MMCCMD writing. The content of this register will be kept after transfer to the shift register. Lower byte of MMCCMD is used for the command index, and upper byte is used for specifying type of command and response, etc. This upper byte is dealt as one pair with the lower byte.

MMCARGL (0003:049A) Offset : 0xD

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ARGL15	ARGL14	ARGL13	ARGL12	ARGL11	ARGL10	ARGL09	ARGL08	ARGL07	ARGL06	ARGL05	ARGL04	ARGL03	ARGL02	ARGL01	ARGL00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	ARGL15- ARGL0	Argument Register. MMCARGL and H is used for specifying the arguments to be sent with the command specified by MMCCMD. Writing to MMCCMD cause start event of sending operation, that is, MMCARG has to be set before writing to MMCCMD. The content of this register will be kept after transfer to the shift register. However, modification to MMCARG is not allowed during sending operation.

MMCARGH (0003:049C) Offset : 0xE

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ARGH15	ARGH14	ARGH13	ARGH12	ARGH11	ARGH10	ARGH09	ARGH08	ARGH07	ARGH06	ARGH05	ARGH04	ARGH03	ARGH02	ARGH01	ARGH00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	ARGH15- ARGH0	Argument Register. MMCARGL and H is used for specifying the arguments to be sent with the command specified by MMCCMD. Writing to MMCCMD cause start event of sending operation, that is, MMCARG has to be set before writing to MMCCMD. The content of this register will be kept after transfer to the shift register. However, modification to MMCARG is not allowed during sending operation.

Note: Bit position of MMCCMD[5-0], MMCARGL and MMCARGH.

MMRs	Bit position of command	
	[47]	start bit
	[46]	transmission bit
MMCCMD[5-0]	[45-40]	command index
MMCARGH	[39-24]	argument
MMCARGL	[23-8]	argument
	[7-1]	CRC7
	[0]	end bit

MMCRSP0 (0003:049E) Offset : 0xF

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MMCRS P015	MMCRS P014	MMCRS P013	MMCRS P012	MMCRS P011	MMCRS P010	MMCRS P009	MMCRS P008	MMCRS P007	MMCRS P006	MMCRS P005	MMCRS P004	MMCRS P003	MMCRS P002	MMCRS P001	MMCRS P000
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	MMCRSP015-000	Response Register0

MMCRSP1 (0003:04A0) Offset : 0x10

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MMCRS P115	MMCRS P114	MMCRS P113	MMCRS P112	MMCRS P111	MMCRS P110	MMCRS P109	MMCRS P108	MMCRS P107	MMCRS P106	MMCRS P105	MMCRS P104	MMCRS P103	MMCRS P102	MMCRS P101	MMCRS P100
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	MMCRSP115-100	Response Register1

MMCRSP2 (0003:04A2) Offset : 0x11

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MMCRS P215	MMCRS P214	MMCRS P213	MMCRS P212	MMCRS P211	MMCRS P210	MMCRS P209	MMCRS P208	MMCRS P207	MMCRS P206	MMCRS P205	MMCRS P204	MMCRS P203	MMCRS P202	MMCRS P201	MMCRS P200
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	MMCRSP215-200	Response Register2

MMCRSP3 (0003:04A4) Offset : 0x12

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MMCRS P315	MMCRS P314	MMCRS P313	MMCRS P312	MMCRS P311	MMCRS P310	MMCRS P309	MMCRS P308	MMCRS P307	MMCRS P306	MMCRS P305	MMCRS P304	MMCRS P303	MMCRS P302	MMCRS P301	MMCRS P300
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	MMCRSP315-300	Response Register3

MMCRSP4 (0003:04A6) Offset : 0x13

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MMCRS P415	MMCRS P414	MMCRS P413	MMCRS P412	MMCRS P411	MMCRS P410	MMCRS P409	MMCRS P408	MMCRS P407	MMCRS P406	MMCRS P405	MMCRS P404	MMCRS P403	MMCRS P402	MMCRS P401	MMCRS P400
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	MMCRSP415-400	Response Register4

MMCRSP5 (0003:04A8) Offset : 0x14

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MMCRS P515	MMCRS P514	MMCRS P513	MMCRS P512	MMCRS P511	MMCRS P510	MMCRS P509	MMCRS P508	MMCRS P507	MMCRS P506	MMCRS P505	MMCRS P504	MMCRS P503	MMCRS P502	MMCRS P501	MMCRS P500
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	MMCRSP515-500	Response Register5

MMCRSP6 (0003:04AA) Offset : 0x15

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MMCRSP615	MMCRSP614	MMCRSP613	MMCRSP612	MMCRSP611	MMCRSP610	MMCRSP609	MMCRSP608	MMCRSP607	MMCRSP606	MMCRSP605	MMCRSP604	MMCRSP603	MMCRSP602	MMCRSP601	MMCRSP600
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	MMCRSP615-600	Response Register6

MMCRSP7 (0003:04AC) Offset : 0x16

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MMCRSP715	MMCRSP714	MMCRSP713	MMCRSP712	MMCRSP711	MMCRSP710	MMCRSP709	MMCRSP708	MMCRSP707	MMCRSP706	MMCRSP705	MMCRSP704	MMCRSP703	MMCRSP702	MMCRSP701	MMCRSP700
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	MMCRSP715-700	Response Register7

Note: Received response is stored in MMCRSP0-MMCRSP7 as the following. MMCRSP0-MMCRSP7 will be updated when receiving, even if the CPU don't have read previous content yet.

In case of R1 or R3 (R4, R5, or R6) response in MMC mode

MMRs	Bit position of command	
MMCCIDX	[47-40]	
MMCRSP7	[39-24]	
MMCRSP6	[23-8]	
MMCRSP5-0	-	

In case of R2 response in MMC mode

MMRs	Bit position of command	
MMCCIDX	[135-128]	
MMCRSP7	[127-112]	
MMCRSP6	[111-96]	
MMCRSP5	[95-80]	
MMCRSP4	[79-64]	
MMCRSP3	[63-48]	
MMCRSP2	[47-32]	
MMCRSP1	[31-16]	
MMCRSP0	[15-0]	

In case of R1 response in SPI mode

MMRs	Bit position of command	
MMCRSP7	[7-0]	It is stored into LSB byte of MMCRSP7
MMCRSP6-0	-	

In case of R2 response in SPI mode

MMRs	Bit position of command	
MMCRSP7	[15-0]	It is stored into LSB byte of MMCRSP7
MMCRSP6-0	-	

In case of R3 response in SPI mode

MMRs	Bit position of command	
MMCRSP7	[39-32]	It is stored into LSB byte of MMCRSP7
MMCRSP6	[31-16]	
MMCRSP5	[15-0]	

MMCRSP4-0	-	
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MMCDRSP (0003:04AE) Offset : 0x17

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	DRSP7	DRSP6	DRSP5	DRSP4	DRSP3	DRSP2	DRSP1	DRSP0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-8	RSV	Reserved Bits. Do not use.
7-0	DRSP7-DRSP0	Data Response The data response byte is stored when it is received in SPI mode. (The CRC status token, consists of three 0 bit, start-bit, CRC status and end-bit, is stored when it is received in MMC mode.)

MMCETOK (0003:04B0) Offset : 0x18

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	ETOK7	ETOK6	ETOK5	ETOK4	ETOK3	ETOK2	ETOK1	ETOK0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-8	RSV	Reserved Bits. Do not use.
7-0	ETOK7- ETOK0	Error Token. The error token byte is stored when it is received in SPI mode.

MMCCIDX (0003:04B2) Offset : 0x19

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	CIDX7	CIDX6	CIDX5	CIDX4	CIDX3	CIDX2	CIDX1	CIDX0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-8	RSV	Reserved Bits. Do not use.
7-0	CIDX7- CIDX0	Command Index Head 1 byte of a response, consists of start bit, transmission bit and command index, is stored when it is received in MMC mode.

MMCKC (0003:04B4) Offset : 0x1A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	CPHA7	CPHA6	CPHA5	CPHA4	CPHA3	CPHA2	CPHA1	CPHA0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-8	RSV	TEST bits. Do not use.
7-0	CPHA7 - CPHA0	CLK start phase control if CPHA = ClockRate(in MMCCLK), when MMCCLK register is set up, MMC clock start immediately. if CPHA < ClockRate when ClockRate - CPHA = N, MMC clock start N*Tcycle(internal function_clock) delay. if CPHA > ClockRate when CPHA - ClockRate = M, MMC clock start (256-M)*Tcycle(internal function_clock) delay.

MMCTORC (0003:04B6) Offset : 0x1B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	TORC7	TORC6	TORC5	TORC4	TORC3	TORC2	TORC1	TORC0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	1	1	1	1	1	1	1	1

Bit	Register Name	Description
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15-8	RSV	Reserved Bits. Do not use.
7-0	TORC7- TORC0	This register shows counter for counting number of blocks to be transferred. Writing to this register is prohibited by User. (Writing function may use for counter test.)

MMCTODC (0003:04B8) Offset : 0x1C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	TODC15	TODC14	TODC13	TODC12	TODC11	TODC10	TODC09	TODC08	TODC07	TODC06	TODC05	TODC04	TODC03	TODC02	TODC01	TODC00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Register Name	Description
15-0	TODC15- TODC0	This register shows counter for counting number of blocks to be transferred. Writing to this register is prohibited by User. (Writing function may use for counter test.)

MMCBLCNC (0003:04BA) Offset : 0x1D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	BLNC7	BLNC6	BLNC5	BLNC4	BLNC3	BLNC2	BLNC1	BLNC0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	1

Bit	Register Name	Description
15-8	RSV	Reserved Bits. Do not use.
7-0	BLNC7-BLNC0	This register shows counter for counting number of blocks to be transferred. Writing to this register is prohibited by User. (Writing function may use for counter test.)

DMATRQ (0003:04BC) Offset : 0x1E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	BRK	TRG
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	W	W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0

Bit	Register Name	Description
15-2	RSV	Reserved Bits. Do not use.
1	BRK	DMA transmission is broken forcibly clear to 0 automatically when hit the BLK bit.
0	TRG	start DDMA transmission clear to 0 automatically when hit the TRG bit.

DMAMODE (0003:04BE) Offset : 0x1F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	DTOIM	DMAM	DIR	DPRMDX	DPRMDR	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV
R/W	-	R/W	R/W	R/W	R/W	R/W	-	-	-	-	-	-	-	-	-	-
Default	-	0	0	0	0	0	-	-	-	-	-	-	-	-	-	-

Bit	Register Name	Description
15	RSV	Reserved Bits. Do not use.
14	DTOIM	Interrupt mask for DMA transfer time_out 0: Interrupt is masked 1: Interrupt is unmasked
13	DMAM	DMA Mode Enable 0: Disable DMA transmission mode. 1: Enable DMA transmission mode.
12	DIR	Direction for DMA transmission 0: MMCIF to SDRAM 1: SDRAM to MMCIF
11	DPRMDX	Word Swap Enable for DMA Write 0: 32bit width data from SDRAM is wrote to card straight. SDRAM_out(31:0) -> CRAD_in(31:0) 1: Data from SDRAM is swapped words by 16bits. SDRAM_out(31:0) -> CRAD_in(15:0,31:16)
10	DPRMDR	Word Swap Enable for DMA Read 0: 32bit width data to SDRAM is read from card straight. CRAD_out(31:0) -> SDRAM_in(31:0) 1: Data to SDRAM is swapped words by 16bits.

		CRAD_out(31:0) -> SDRAM_in(15:0,31:16)
9-0	RSV	Reserved Bits. Do not use.

DMASTAD0 (0003:04C0) Offset : 0x20

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	TAD15	TAD14	TAD13	TAD12	TAD11	TAD10	TAD09	TAD08	TAD07	TAD06	TAD05	TAD04	TAD03	TAD02	TAD01	TAD00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	TAD15 - TAD0	Lsb 16bits of DMA start address. (Relative address)

DMASTAD1 (0003:04C2) Offset : 0x21

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	TAD26	TAD25	TAD24	TAD23	TAD22	TAD21	TAD20	TAD19	TAD18	TAD17	TAD16
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-11	RSV	Reserved Bits. Do not use.
10-0	TAD26 - TAD16	Msb 10bits of DMA start address. (Relative address)

DMASTAT0 (0003:04C4) Offset : 0x22

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	REST15	REST14	REST13	REST12	REST11	REST10	REST09	REST08	REST07	REST06	REST05	REST04	REST03	REST02	REST01	REST00
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-0	REST15-REST0	Lsb 16bits of transfer byte rest

DMASTAT1(0003:04C6) Offset : 0x23

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	TOUTDT	RUN	REST27	REST26	REST25	REST24	REST23	REST22	REST21	REST20	REST19	REST18	REST17	REST16
R/W	-	-	R	R	R	R	R	R	R	R	R	R	R	R	R	R
Default	-	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-14	RSV	Reserved Bits. Do not use.
13	TOUTDT	Time-Out (DMA transfer acknowledge) 0: No time-out occurs. 1: Timed-out has occurred during waiting acknowledge from SDRAMC.
12	RUN	DMA transmission flag 0: finish transmit. 1: DMA transmission is running.
11-0	REST27 - 16	Msb 12bits of transfer byte rest

DMATOR (0003:04C8) Offset : 0x24

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	DTPR7	DTPR6	DTPR5	DTPR4	DTPR3	DTPR2	DTPR1	DTPR0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bit	Register Name	Description
15-8	RSV	Reserved Bits. Do not use.
7-0	DTOR7 - 0	Time-Out Count for DMA transfer acknowledge from SDRAM DTOR = 0000 0000 0000 0000 no time-out DTOR = 0000 0000 0000 0001 1 CLK clock DTOR = 0000 0000 0000 0010 2 CLK clocks : DTOR = 1111 1111 1111 1111 65535 CLK clocks

Registers specific to SDIO card

SDIOCTL (0003:04CA) Offset : 0x25

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RDWTCR	RDWTRQ
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0

Bit	Register Name	Description
15-2	RSV	Reserved Bits. Do not use.
1	RDWTCR	Read Wait enable for CRC error. If this bit is set, Read Wait operation automatically starts in case a CRC error is detected during multiple block read access, and while it's not the last block to be transferred. In this case, RDWTRQ (Read Wait request) is automatically set. To end the Read Wait operation, write '0' to RDWTRQ. (No need to clear RDWTCR.)
0	RDWTRQ	Read Wait request. 0: End Read Wait operation and release DAT[2]. 1: Set a Read Wait request. If this bit is set, Read Wait operation starts 2 clocks after the end of the read data block. MMCIF asserts low level on DAT[2] until this bit is cleared.

SDIOST0 (0003:04CC) Offset : 0x26

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RDWTST	INTPRD	DAT1
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	R	R	R
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0	1

Bit	Register Name	Description
15-3	RSV	Reserved Bits. Do not use.
2	RDWTST	Read Wait status. RDWTST becomes '1' during Read Wait operation.
1	INTPRD	Interrupt period. INTPRD becomes '1' during Interrupt period.
0	DAT1	DAT[1] status. DAT[1] can be monitored through this bit.

SDIOIEN (0003:04CE) Offset : 0x27

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RWSEN	IOINTEN
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0

Bit	Register Name	Description
15-2	RSV	Reserved Bits. Do not use.
1	RWSEN	Read Wait interrupt enable. 0: Disable the interrupt. 1: Enable the interrupt. Interrupt is generated to notify that Read Wait operation has been started.
0	IOINTEN	SDIO card interrupt enable. 0: Disable the interrupt. 1: Enable the interrupt.

SDIOIST (0003:04D0) Offset : 0x28

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RWS	IOINT
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0

Bit	Register Name	Description
15-2	RSV	Reserved Bits. Do not use.

1	RWS	Read Wait interrupt status. 0: Not interrupted. 1: Interrupted. This bit is set to '1' when Read Wait operation starts and Read Wait interrupt is enabled. To clear this bit, write '1'.
0	IOINT	SDIO card interrupt status 0: Not interrupted. 1: Interrupted. This bit is set to '1' if the SDIO card interrupt is detected and card interrupt is enabled. To clear this bit, write '1' after the interrupt on the DAT[1] line is removed.